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ABSTRACT

We give a survey on the history of computers in Japan that have been produced over the last 50 years. Here, we focus on the topics of personal computers (PCs). We can see PCs produced in Japan have very unique features. They coevolved with some other related machines, e.g. word processing machines, game machines, etc. They also had several localised features such as capability of dealing with Japanese language. At the same time, developers of PCs struggled to find a good solution between localisation and globalisation. In the early stage of the history of PCs, their architectures were all different, and software had to be developed and tuned for each of them. Thus providing a common platform to all PCs on which every software can run was indispensable for globalisation. In this paper, how PCs in Japan have been evolved and how such problems were attacked are discussed based on the author's experiences of developing early PCs.



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1. Introduction

A computer is a product that cannot be completed only with the computer itself. It is unusable without software. In addition, peripheral devices such as printers and storage devices are necessary. Those software and peripheral devices are provided by third-party manufacturers other than the manufacturer of the computer itself. Because the third-party manufacturers target the best-selling computer to maximise their sales, such a computer gets better and cheaper software and peripheral devices than other computers. By this, it sells much more. This positive feedback mechanism makes the

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best-selling computer dominates the market. In order to compete with the top selling manufacturer, it becomes effective for the second and subsequent computer manufacturers to provide products that are compatible with those from the top manufacturer. This is because they can use the software and peripherals provided by the third-party manufacturers.

For the past 50 years, computer makers with the largest market continued to be U.S. makers, who have been expanding their business based on the big U.S. market. Therefore, Japanese computer makers inevitably decided to adopt the technology of U.S. manufacturers and took a strategy to develop compatible products to survive. However, even in these trends, there were other activities in the manufacturers in Japan. They not only produced U.S. compatible computers but also developed computers that are useful for Japanese users. In particular, by providing excellent Japanese language support, which was not possible by the U.S.-made machines, they had gained a large share in the Japanese market. On the other hand, there was also an activity of creating a world standard computer such as MSX personal computers (see Section 6) for general consumers and selling it all over the world. Such a project was carried out based on the technical strengths of the Japanese consumer electronics industry.

In the latter half of the 1970s, the beginning of the market introduction of personal computers, I was involved in this industry. Since then, I had been engaged in introducing U.S. technology into Japan, developing systems for Japanese language support, and creating a standard personal computer for general consumers. Based on these experiences, I will explain how personal computers have penetrated into the Japanese society.

Contents of the following sections are as follows. Section 2 gives a brief history of computers in Japan before personal computers. Sections 3–8 explain several important achievements in the history of personal computers and related products. Sections 9 and 10 describe some on-going activities other than personal computers. Section 11 gives concluding remarks of the author.

2. Before personal computer

The history of computers in Japan before personal computers has the following two stages:

1. The stage of developing Japan's original prototype and commercial computers.
2. The stage of developing IBM mainframe compatible machines.

In this section, we briefly explain these stages and how government supported their developments, see also [1,2].

2.1. Original computers in Japan in the early stage of the history

The history of computer development in Japan began with a computer using relay device, next vacuum tubes, and then transistors as in other countries. It is remarkable that a very unique device called the parametron[1] also existed. It was invented by Eiichi Goto of the University of Tokyo in 1954. Since the parametron is more reliable than a vacuum tube, it was adopted by several domestic computer systems. However, it could not compete with a transistor in the performance, and thus ceased to be used in the 1960s.

In the early computer development in Japan, there were two lines of development activities. One is by each computer manufactures that developed their own systems. Another activity was under the initiative of a national telecommunications company called Nippon Telegraph and Telephone Public Corporation.

2.2. Mainframe computers compatible with the IBM machine

The Japanese government considered the mainframe computers to be important as the core system of the information industry, and thus tried to prevent the industry from being monopolised by U.S.

companies. As a result, national level support was provided to develop alliances with U.S. companies and promote domestic system development. Each Japanese manufacturer has developed products by introducing technology from a U.S. company who had been succeeding in a much larger market in the U.S.A. by signing a license agreement as listed in the following.

NEC	Honeywell (1963)
Toshiba	GE (1964)
Mitsubishi Electric	TRW (1963)
Oki Electric (Oki Univac)	Univac (1963)
Hitachi	RCA (1965)
Fujitsu	Amdahl (1974)

In 1971, the Japanese government changed its previous protection trade policy and reorganised the industry to liberalise the import of computers. It compiled computer manufacturers into three groups: Fujitsu/Hitachi, Mitsubishi Electric/Oki Electric and NEC/Toshiba, and promoted those groups by providing approximately 57 billion yen subsidy in the 5 years from 1972 to 1976 (see Table 1).

Fujitsu had continued the development focusing on proprietary technology, however, in 1974, converted to an IBM plug compatible manufacturer by investing in Amdahl Inc., which had been developing such computers. As a result, the Fujitsu/Hitachi group succeeded to provide good IBM plug compatible computers. They were welcomed by the market by offering these machines with better cost performance. However, the mainframe computer market has been shrinking due to the spread of personal computers (Figure 1).

2.3. Support of computer industry by the government

As the national projects, the government had implemented the following development projects in order to promote the computer industry. However, none had succeeded except for the Super LSI Development Project.

- Super high-performance computer development project (1966–1972) (budget 10 billion yen): It tried to develop a standard architecture among Japanese manufacturers, but it did not go well.
- Super LSI Development Project (1976–1979) (budget 70 billion yen): It developed semiconductor production technology. It contributed to the success of the DRAM semiconductor industry later.
- Fifth -Generation Computer (1979–1992) (budget: 57 billion yen): It aimed to develop artificial intelligence, but it was premature, with no remarkable achievement.
- Sigma Project (1985–1990) (budget: 25 billion yen): In line with the spread of UNIX OS, it aimed to develop and standardise Japanese UNIX tools, but was not able to achieve results.
- Information Navigation Project (2007–2009) (budget: 15 billion yen): In order to counter the monopoly of search engines by Google, it aimed to develop information retrieval and analysis technology and applied technology, but it could not achieve remarkable results.

Table 1. Chronological table of mainframe computer events.

Year	Mainframe Computer events
1952	IBM 701 released (IBM's first commercial computer)
1959	IBM 7090 released (first transistor computer)
1960	Japanese computer makers made a patent contract with IBM
1964	IBM System/360 released
1966	VHSCS (Very High-Speed Computer System) project started
1970	IBM SYSTEM 370 released
1972	3 Technical Research Cooperatives formed by: Fujitsu/Hitachi, Toshiba/NEC, Oki/Mitsubishi
1974	M/ACOS/COSMO Series released as the result of technical research cooperatives
1975	Amdahl 470/V6 Fujitsu M-190 released (IBM plug compatibles)

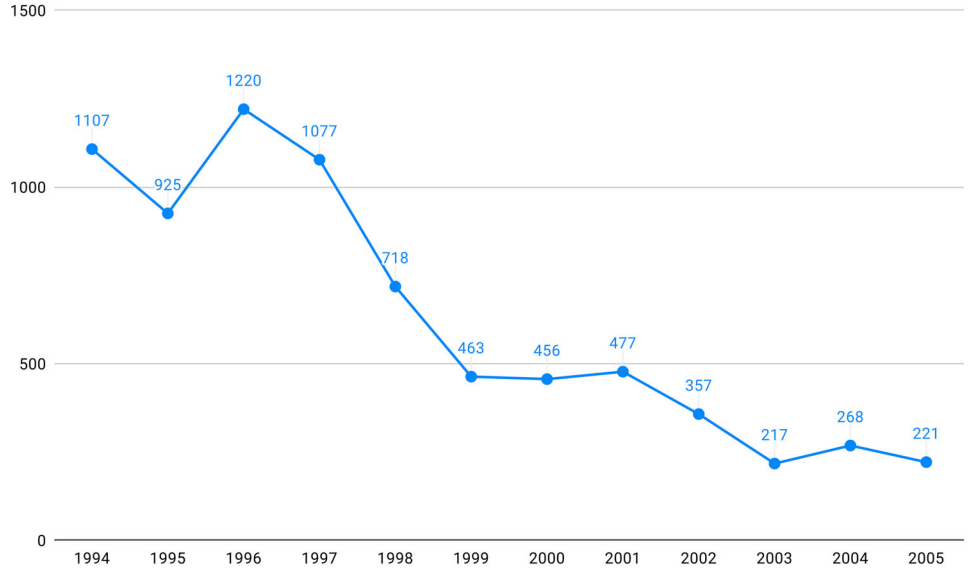


Figure 1. Decline of domestic shipments of large mainframe computers with a price of 250 million yen or more (source: Japan Electronics and Information Technology Industries Association).

3. History of personal computers

In the latter half of 1970s, the personal computer era started. In this section, we give a short survey of the history of them. We then explain activities of ASCII Corporation, who played important roles in the development of early personal computers in Japan.

3.1. Short survey of the personal computer history

In the early stage of the development, personal computers were too incompetent compared to the mainframe computer. Hence, the importance of the personal computers was never recognised by the computer industry nor by the government. It is symbolised by the fact that initially the division that promoted the personal computers was that of semiconductors rather than the department of computers even in a computer manufacturer. Later, when the performance was improved to the level that a personal computer can be used for business, and in particular when IBM released a personal computer for business, each Japanese computer maker also began to develop full-scale personal computers. However, unlike the case of the mainframe computers, there was little support at the government level. Table 2 shows chronological table of personal computer events. See also [3,4] for the history of personal computers in Japan.

As Japan's technical capabilities are quite high, sometimes products based on Japanese original technology are developed and released to the Japanese market. However, a computer that acquired the largest market gets benefit from abundant supply of application software and peripheral devices by third-party manufacturers. Thus the computer further expands its market share. As a result, the Japanese market was dominated by the hardware and software originated in the U.S.A. that succeeded in the global market, and Japan's original systems could not survive. It is often called the Galapagos syndrome[2] in Japan, i.e. various products made their own unique evolution in Japan, like the creatures in the Galapagos Islands. However, they could not adapt to the global market and finally disappeared. This is because the computer market is globally connected, unlike the Galapagos Islands.

There was also an important activity in developing an operating system called TRON,[3] which is an acronym for The Real-time Operating system Nucleus, and was proposed by Dr. Ken Sakamura of the

Table 2. Chronological table of personal computer events.

Year	Personal Computer events
1973	Intel 8080 CPU chip released
1974	Altair 8800 released
1975	Microsoft established
1977	Apple II released
	ASCII corporation established
1979	NEC PC-8001 released
1981	IBM PC released
	NEC PC-9801 released
1983	Radio Shack TRS-80 M100 released
	MSX released
1984	Apple Macintosh released
	IBM PC/AT released
	Chips & Technologies established
1985	Windows 1.01 released
	MSX2 released
1987	AX consortium established
1988	MSX2+ released
1990	DOS/V released
	MSX turbo R released
1991	OADG consortium established
1992	Windows 3.1 released
1994	Netscape Navigator 1.0 released
1995	Windows 95 released

University of Tokyo at that time. The TRON project started in 1984 and was greeted favourably by the Japanese computer society (see also Section 9). However, in 1980s it could not continue the promotion policy by the pressure due to the trade friction between Japan and the U.S.A.

3.2. Roles played by ASCII Corporation[4]

ASCII Corporation was founded in 1977 as a publisher that published the magazine 'ASCII' which was a comprehensive microcomputer magazine. ASCII Corporation and one of its founder Kazuhiko Nishi had a great influence on the history of Japanese personal computers. The first major activity was the establishment of partnership with Microsoft Corporation. In 1978, Nishi decided to act as Microsoft's Japanese agent in negotiations with Bill Gates of Microsoft. He introduced Microsoft's system software to personal computers of almost all major Japanese manufacturers. Since he succeeded to sell Microsoft products into Japanese industry, established ASCII Microsoft Corporation, and contributed the growth of Microsoft as a vice president of Microsoft Corp.

ASCII Corp. not only promoted the distribution of Microsoft's software into Japan as an agent of Microsoft but also made a significant contribution in the area of personal computer technology. It contributed to the development of Shift JIS code, which is a character code system with a very little overhead of Japanese language processing. By this, the efficiency of processing of Japanese in the U.S.-made software and the localisation of Microsoft software were greatly improved. It also contributed to the Japanese localisation of UNIX operating system as well. In addition, it developed the standard of personal computer called MSX (see Section 6). It was produced by 13 Japanese manufacturers and 8 overseas manufacturers, and distributed to various countries worldwide. I was in a cooperative relationship from the inception of ASCII Corp. Since joining in 1982, I had been a responsible person for the above-mentioned technological development. So, in the following, I will describe the history of personal computers in Japan based on my experiences.

Introduction of Microsoft's software technology into Japan was done by sending engineers including the author from ASCII Corp. to Microsoft. Those engineers acquired skills by developing products for Japan, and later brought the technology back to Japan. The first major achievement was a



Figure 2. NEC PC-8001 Personal Computer.

development of the laptop computer sold by Radio Shack, NEC and Olivetti, that was manufactured by Kyocera (see Section 4.2).

4. Early personal computers and laptop computers

Here, we observe how very early personal and laptop computers were produced in Japan.

4.1. Personal computer NEC PC-8001

PC-8001 personal computer (of PC-8000 series) made by NEC in 1979 was a full-scale personal computer and the first popular machine in Japan (Figure 2). It had Microsoft's system software (BASIC interpreter) installed. Regarding the system software, NEC had a product developed by themselves, but it was recognised by a persuasion from ASCII Corp., that it is important to adopt the global standard. Thus NEC stopped using its own software and adopted the software from Microsoft. Since this personal computer made by NEC was successful, it became the catalyst to use Microsoft's software by each subsequent Japanese manufacturer, and Microsoft got a great success in Japanese market.

4.2. Laptop computers M100/M10/PC-8200

Early laptop computers appeared in the former half of 1980s. Figure 3 shows a TRS-80 M100[5] laptop computer sold by Radio Shack in 1983. This computer was born from a conversation between Nishi of ASCII Corp. and Kazuo Inamori, President of Kyocera at that time, who were accidently present on the same airplane connecting the U.S.A. and Japan. Its hardware was developed by Kyocera, and its software was developed by three engineers including the author dispatched from ASCII Corp. to Microsoft Seattle office.

This computer received a great success because of its ease of use, and it was said that 6 million units were sold worldwide. Since it had a built-in communication modem and a basic function of connecting to a network, it was particularly welcomed and popularised by journalists who wrote articles and sent them using the built-in modem. Due to Nishi's and Kyocera's sales efforts, this computer was sold not only by Radio Shack but also by NEC (PC-8201) in Japan and Olivetti in Italy (M10) (Figure 4). At that time, ASCII's engineers had reached the level to acquire Microsoft's technology and introduce it to Japanese personal computer industry in cooperation with Microsoft.



Figure 3. Radio Shack TRS-80 M100 laptop computer (source: https://commons.wikimedia.org/w/index.php?curid = 3378530 (PD)).



Figure 4. NEC PC-8201 (left), and Olivetti M10 (right) (source: https://en.wikipedia.org/wiki/TRS-80_Model_100#/media/File:NEC_PC-8201_001_JPN.jpg (CC BY-SA 3.0) https://it.wikipedia.org/wiki/Olivetti_M10#/media/File:Sottsass05.JPG (CC BY 2.5)).

5. Japanese language support

In introducing a global standard computer into Japan, there is a field in which Japan's unique system is absolutely necessary. It is Japanese language support. In addition to the alphanumeric characters defined by the ASCII code of the U.S.A., Japanese characters have the following character sets.

- Kana characters (katakana and hiragana)
- Kanji characters.

Kana characters are phonetic ones indicating pronunciations, and there are two-character sets, katakana and hiragana. These two sets represent the same set of pronunciations, such as upper case letters and lower case letters of the Roman alphabet. Katakana is mainly used for expressing names of species or words that came from foreign languages, while hiragana is used for all other cases. Katakana has been supported since the era of the mainframe computers because its shape is simple and can be expressed with a small number of pixels. Kanji characters are the ones originally invented in China. In Unicode, 87,882 Chinese characters are registered, while about 7000 kanji characters are used in Japanese computers. Since the shape of kanji is complicated compared with Roman characters, katakana, and hiragana, at least 16 dots × 16 dots are required which is four times as large as the alphanumeric character which can be displayed with 8 dots × 8 dots. In the early stage of the development of personal computers, to display kanji with the speed comparable to that of alphanumeric characters, it was necessary to have hardware support, which has become a barrier to the introduction of the global standard personal computer into the Japanese market.

5.1. Input/output method for Japanese

In this section, we give a survey on the input/output methods for Japanese employed in the early Japanese computers.

5.1.1. Input of kana characters

Since the number of kana characters is not many, they can be accommodated in the alphanumeric keyboard, as shown in Figure 5, by overlaying kana characters on the QWERTY keys. Kana input is done by switching the input mode. Later, the method of entering the reading (pronunciation) of the kana character by the Roman alphabet keys has become popular (a conversion method of Roman characters to kana characters). This is because it only needs to remember the arrangement of the Roman alphabet keys.

5.1.2. Input of kanji characters

5.1.2.1. Japanese typewriter style kanji keyboard[6]. As a method of entering kanji which consists of thousands of characters, there was a method of selecting kanji characters directly using the Japanese typewriter style keyboard which had a large two-dimensional keyboard. However, since such a keyboard is expensive and not convenient, it was never used with a personal computer featuring low price, except for some office computers.



Figure 5. Japanese keyboard layout (source: https://commons.wikimedia.org/wiki/File:109keyboard.svg (PD)).

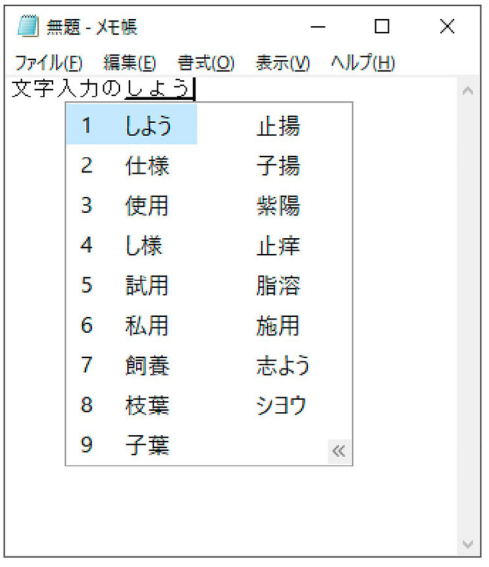


Figure 6. Japanese kana–kanji conversion.

5.1.2.2. Single kanji conversion. Instead of using kanji keyboard, a method of inputting kanji readings with kana characters on a normal QWERTY keyboard and converting it into a kanji character was used in the early systems. Since each kanji character has multiple readings, and for each reading there correspond multiple kanji characters, it is necessary to have a dictionary that records the correspondence between kanji characters and readings. Then, by referring this dictionary, it is possible to select a kanji character from the displayed candidates one by one. However, since one can convert only one kanji character at a time by this method, it is inefficient and difficult to use. However, early computers could use only this method because the conversion dictionary could not be enlarged due to the memory size constraints.

5.1.2.3. Phrase conversion. A method of recognising phrases (Bunsetsu) in a given kana character string in accordance with Japanese grammar, and converting them into kanji and kana character strings has been developed and used until now. Several kinds of such kana–kanji conversion systems are provided as a subsystem of an operating system called IM (Input Method). Although conversion algorithms are being sophisticated, there is a problem that Japanese has many different words having the same reading. IM needs to display the most probable kana/kanji strings from the reading, and the user has to choose the appropriate candidate from the proposed menu to input the desired Japanese sentence. In the example shown in Figure 6, a total of 17 candidates are presented by IM. The user's selection is learned by the IM, and when the same reading is input next time, the last selected character string is presented as the first candidate. However, in order to optimise the selection operation, the user needs to input a character string considering the status of priority order of the conversion dictionary.

When entering sentences composed only of alphanumeric characters, it is almost unconsciously and smoothly performed to transfer the sentences that came to one's mind into the computer system through fingers. However, when entering Japanese sentences, it is necessary to input them while worrying about the state of IM. Hence, it is considered to be a great disadvantage for Japanese people in using computer. We hope, in the near future, advancement of AI technology will improve this problem.

5.1.3. Output of kanji characters

5.1.3.1. Display system. In the early personal computers, alphanumeric characters were shown on the display in 8 dots × 8 dots, and the display resolution of 640 × 200 dots were used to show 80 characters by 25 lines (such as in IBM PC CGA). On the other hand, kanji characters with complex shapes required 16 dots × 16 dots to show, which have a double width of that of alphanumeric characters. Also, in order to display 25 lines of characters the same as alphanumeric characters, the vertical resolution of the display need to have 400 dots which was twice of the resolution of the U.S. system.

5.1.3.2. Printer. Initially, dot matrix printers which print kanji character with 16 dots × 16 dots, the same resolution as that of displays, were used. But, later, printers which print kanji with 24 dots × 24 dots were developed using a relatively inexpensive thermal head, and they were widely used.

5.2. Character code for Japanese

The number of alphanumeric characters contained in the ASCII code, which is a character code used in the U.S. personal computer, is 94 excluding the blank character and is represented by the 7-bit code. The JIS X 0208 specification[7] is a kanji character code standard defined in 1978 by the Japanese Industrial Standard (JIS). It defines 6879 kanji characters in the table of 94 rows and 94 columns, and thus makes it possible to represent a kanji character by a combination of two 7-bit character codes. Therefore, it is necessary to distinguish the kanji character represented by two 7-bit codes from the alphanumeric characters each of which is expressed by one 7-bit code. There are two possible methods for this.

One method is to switch the alphanumeric characters and the kanji characters by inserting an escape sequence before a character string to indicate which character set it is now used. There is a merit to use this method, since kanji character codes can be transmitted on a 7-bit transmission line. Therefore, this method is still used until now in the e-mail message transfer.

Another way to distinguish between alphanumeric characters and kanji characters is using 8 bits to represent a character code, i.e. setting the most significant bit (i.e. the 8th bit) to 0 or 1. However, at this time in Japan, katakana characters were mapped in the space defined by the 8th bit as 1 in the scheme defined by the JIS X 201 specification,[8] and this scheme was used in mainframe computers and office computers. Therefore, if we simply map kanji character codes represented by two 7-bit codes defined in JIS X 208 in a space whose 8th bit is 1, they collide with the katakana character codes defined by the specification of JIS X 201 since the code regions overlap. Thus it becomes impossible to make JIS X 208 kanji and JIS X 201 katakana characters coexist in this scheme.

5.3. Shift JIS code

Since IBM entered the personal computer business in 1981, Japanese computer makers also got the momentum to develop full-fledged personal computers for business use. In developing a personal computer for business in 1981, Mitsubishi Electric planned to procure applications such as language processing systems and spreadsheets from Microsoft, and an operating system from Digital Research. In order to fully support Japanese language for business use, Mitsubishi Electric requested to discuss with Digital Research and Microsoft on the handling of Japanese character codes.

At that time, I was engaged in the work of porting Microsoft's software to the personal computers of several Japanese manufacturers at the Microsoft office. As the only engineer with Japanese language knowledge at Microsoft, I was in charge of handling Japanese character codes. I visited Digital Research and conducted discussions, but Digital Research had no engineer with sufficient knowledge of Japanese, and from then on, I was in charge of designing the character code for personal computers to use in Japanese market.

We decided to satisfy the following three requirements as the design goal for the Japanese character code used on the personal computer.

1. Minimise the amount of program change required for making U.S.-made software be able to handle Japanese (i.e. make U.S.-made software handle Japanese at the minimum cost)
2. Minimise the processing overhead required to handle Japanese (i.e. realise a performance which is not much different from the performance of processing only of alphanumeric characters even by a poor computing power)
3. Can coexist with the JIS X 201 8-bit kana character code (i.e. can use existing data asset without modification).

In order to satisfy the conditions 1 and 2, it was effective to make the number of characters proportional to the length of the character code data that express those characters. The software of the personal computer has to update the characters displayed on the screen in real time according to the operation of the user. A kanji character expressed in 2 bytes is displayed on the screen with the double width of an alphanumeric character represented by 1 byte. Therefore, if the alphanumeric character is represented by 1-byte code and the kanji character is represented by 2-byte code, the strain on the U.S. program is the smallest and the processing software overhead can be minimised to support kanji characters on the display screen. If we adopt the method of inserting the escape sequence, then the data length of the character codes becomes different from the number of characters. Therefore, it is preferable to choose the method of expressing the 2-byte kanji code by setting the 8th bit as 1.

Regarding the requirement 3, I did not place much emphasis on it initially, but I had a discussion with Hiroyuki Fukase of ASCII Microsoft Corporation. He was communicating closely with Mitsubishi Electric in Japan and had an experience in the field of mainframe and office computers. As a result of the discussion, we had concluded that it is important as a character code of a new era to make it possible to handle vast amounts of data created and accumulated in existing JIS X 201 without modification. Therefore, we decided to adopt the method of allocating JIS X 208 code using vacant space that JIS X 201 is not using, whose basic idea was proposed by Mitsubishi Electric.

This was the beginning of the coding scheme for Japanese characters called Shift JIS code,[9] which was adopted by all Japanese personal computers starting with Mitsubishi Electric's Multi-16[10] personal computer shipped in 1982. This character coding scheme was widely adopted in the products other than personal computers, such as mobile phones. Since Shift JIS coding scheme keeps the relation that a kanji code is in 2 bytes and alphanumeric and kana codes are in 1 byte, it becomes easy for U.S.-made software to handle Japanese language. The fact it can avoid complicated handling of escape sequences also simplified the software development work. These are the reasons why Shift JIS code was supported and spread widely.

Figure 7 shows how JIS X 0208 codes are mapped into the space of 2-byte codes while coexisting with JIS X 201 codes. The first byte of the Shift JIS code that consists of 2 bytes is mapped to 81 to 9F, E0 to EF area while avoiding the area of A0 to DF used by JIS X 201. The area from F0 to FC was defined as an area where manufacturers can define their own characters. Each manufacturer independently defined and used various characters which are not given by JIS, and thus an area for accommodating those unique characters was required.

In this way, the first byte of the Shift JIS code is allocated to an area not used by JIS X 201, so there is no possibility of conflict. However, the second byte uses the area used by JIS X 201 because the capacity of the area not used by JIS X 201 was not sufficient. When software processes character codes, no problem arises if it recognises and processes whether each character code byte is the first byte or the second byte of Shift JIS code. However, we cannot expect all U.S. software to do such processing, in reality.

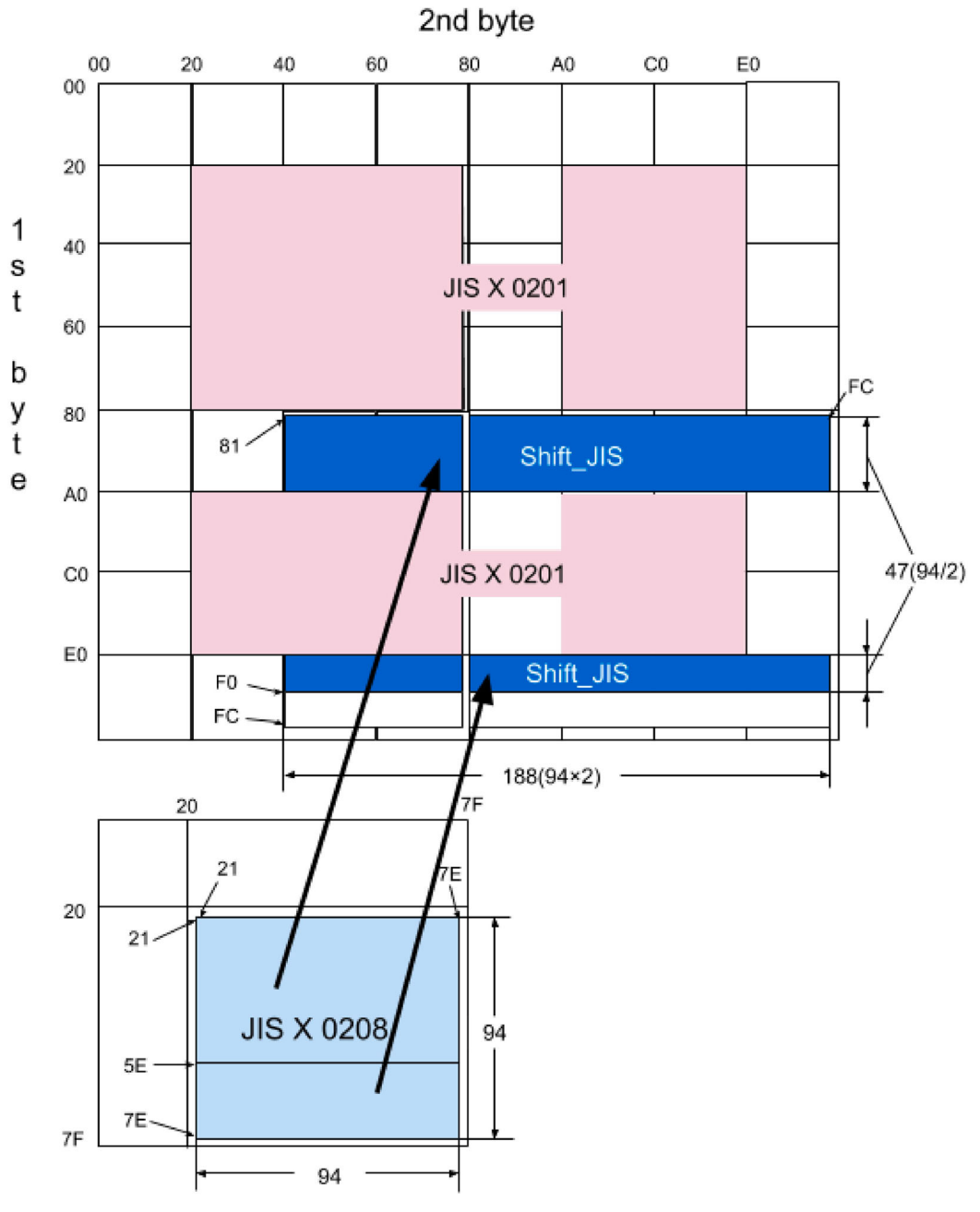


Figure 7. Code mapping scheme of Shift JIS code.

There are characters called meta-characters[11] that represent special functions. For example, a double quotation mark (") is used to indicate the beginning and end of a string. Some programs that process character data treat characters surrounded by two double quotation marks as one character string. In software that does not recognise 2-byte character code, when the double quotation mark character code appears in the second byte of the Shift JIS code, it is regarded as the end of the character string, and thus the character string cannot be correctly recognised. To avoid problems as much as possible with the programs that do not perform 2-byte code handling, the second byte of shift JIS

codes is mapped from 40 to avoid using character codes existing in 20 to 3F area that are used as various meta-characters.

The following symbols reside in 20 to 3F area, excluding numerals.

!	"	#	$	%	&	'	(	)	*	+	,	−	.	/	:
;	<	=	>	?											

However, it was not possible to avoid using character codes of following symbol characters of 40 to 7F that contain meta-characters in the second byte of the shift JIS code.

@	[	\	]	^	_	'	{	|	}	~

Of these inevitable characters, 5C (backslash) is used as a special meta-character in the following cases. When this character appears as the second byte of the Shift JIS code, it causes the following problems in the processing systems which do not process 2-byte character codes.

1. Since it is used as a file path delimiter in MS-DOS/Windows, Japanese path names may not be interpreted correctly.
2. Since it is defined as an escape character in HTML and many other language processing systems, characters that contain 5C in the second byte are handled incorrectly because they are treated as escape meta-character instead of a 2-byte character.

The Shift JIS coding scheme given here was designed to efficiently process kanji characters within the processing system which handles the 2-byte character code properly, and it was not supposed to be used for exchanging data between systems. However, because it can coexist with ASCII code and JIS X 201, it was brought out of the system and used to exchange information, causing the above problems.

Shift JIS code was defined in Annex 1 of JIS X 208 in 1997, because it was widely used.

The standard specifying the Japanese character code used in today's Japanese computer is JIS X 0213[12] that defines 11,233 characters in total, and it also has interoperability with the Unicode character code system. This is an extension of JIS X 0208 which defined 6879 characters.

There is another coding method called EUC[13] (Extended UNIX Code) for Japanese characters. It was defined in 1985 and is widely used on UNIX computers. This coding scheme maps JIS X 208 as it is with the 8th bit set to 1. The 1-byte katakana character of JIS X 201, where the code area conflicts, moved to the 2-byte code area prefixed with 8E. When comparing the EUC code with the Shift JIS code, it could avoid the problem that ASCII code characters appear in the 2nd byte of the Shift JIS code, so there is an advantage that the problem like the above 5C problem does not occur. However, the 1-byte katakana character codes of JIS X 201 cannot be handled directly.

6. MSX computers[14]

MSX is a standardised architecture of a personal computer developed in ASCII Microsoft in 1983, and manufactured by many companies (Figure 8). In this section, we discuss how it was developed, how its features are, and how it influenced computer societies in Japan and foreign countries.

6.1. Overview of MSX

The history of computers has been built with rapidly advancing hardware, and software that developed manually and labour intensively to utilise the hardware. Development of software requires a



Figure 8. MSX computers from various manufacturers (source: Weekly ASCII Magazine).

lot of manpower, which is time-consuming and expensive. No matter how good hardware is developed, it is not usable without software that drives that hardware. Therefore, computer systems must be developed considering how easy it is to develop and maintain software. However, early personal computer makers considered the processing system of BASIC language, which was installed as standard function of the personal computer, as a target of differentiation, i.e. an item to differentiate their products to compete with other companies. Each company had asked Microsoft to develop a language processing system with better language specification than the other preceding competitors.

At that time Microsoft had got orders to provide the BASIC language processing system from the majority of Japanese personal computer makers. As a result, new products equipped with processing systems of different language specifications were introduced into the market. Even if language processing systems of those new product are made by the same Microsoft, the specifications are different from those of the past. Hence, it was not possible to operate a program developed so far. It was necessary to do a barren work called 'porting' to modify the existing software according to the 'improved' language specification. Basically, every effort of software engineers should be directed to the development of new software that creates a new value. I felt it very resentful for engineers to consume their manpower with so-called 'porting' work that does not produce any new value.

Software requires a lot of efforts to design, develop, debug, and make it stable. Also, even if it becomes stable once, it is necessary to make maintenance modifications in response to changes in the usage environment. Thus when developing a computer system, it should be considered with the top priority how we develop expensive and time-consuming software for it. I thought it is not a right way that manufacturers demand different language specifications for each of new personal computer models. Therefore, I strongly insisted my opinion to Vice President Nishi of Microsoft and ASCII Corp. that the development of BASIC should not be continued in this way. In response to my opinion, Nishi proposed the concept of MSX computer.

At that time, personal computer applications were spreading to business applications by the introduction of a 16-bit CPU, and the main battlefield of personal computer business had shifted to a model using a 16-bit CPU as symbolised by the entry of IBM. Therefore, the system using an 8-bit CPU was no longer in the mainstream of development even in Microsoft. As a result, it became possible to develop 8-bit personal computers led by ASCII Corp. using Microsoft's software assets. The place of development was located at the ASCII's office in Tokyo by installing the same mainframe computer (DEC SYSTEM 20) for development as in Microsoft. In 1983 we released MSX system with the approval of leading home electronics manufacturers. Since then, I had served as development director of the MSX system for 10 years. The personal computers of MSX standard were released by the following 13 Japanese manufacturers and 8 overseas manufacturers.

Domestic:

Canon Inc.	CASIO COMPUTER CO., LTD.
FUJITSU LIMITED	FUJITSU GENERAL LIMITED
Hitachi, Ltd.	Panasonic Corporation
Mitsubishi Electric Corporation	Yamaha Corporation
Victor Company of Japan, Limited	PIONEER CORPORATION
SANYO Electric Co., Ltd.	Sony Corporation
TOSHIBA CORPORATION	

Overseas:

Royal Philips (Nederland)
SPECTRAVIDEO INTERNATIONAL LIMITED (USA)
Telematica (Argentina)
Gradiente (Brazil)
Sharp de Brazil (Brazil)
Goldstar (Korea)
Daewoo (Korea)
Samsung (Korea)

The price tag of NEC's PC-9801, the representative of a16-bit personal computer at that time, was 298,000 yen for the main unit alone, which was too expensive for general consumers. On the other hand, MSX computer's price tag was around 50,000 yen. In addition, since a normal television can be used as a display, there is no need to buy a display. So, anyone could purchase it. Though it was a low price product, we took full advantage of the know-how of developing personal computers at Microsoft and designed it as a full-fledged computer that had sufficient functions, performance, and expandability as an introductory computer.

We made big efforts to maintain compatibility of the machines that allows the same application software and the same peripheral device to run even if the manufacturers are different. Although compatibility was realised by the same system software built on the same hardware, we also recognised that it was necessary for manufacturers as business requirements to be able to make products with distinctive features to compete with each other. That is flexibility of the machine. In order to satisfy such contradictory conditions of full compatibility and flexibility required by the manufacturers, the architecture of the system bus called a 'slot' is devised.

6.2. Slot-bus architecture of MSX

A slot is an opening for inserting a cartridge. Since the system bus is provided in this slot, any device can be incorporated into the system by connecting it in the slot. In the MSX standard, it was obliged to provide at least two slots in the computer main body. Users are able to expand the system by inserting



Figure 9. MSX software cartridges.



Figure 10. MSX MODEM cartridge (courtesy of ABIT corp.).

various cartridges into the slot such as a ROM cartridge containing a game software (Figure 9) or a modem cartridge for communication (Figure 10).

Since slots are arranged in the address space separated by the paging technique, the addresses of memories, peripheral devices and others arranged in the multiple slots do not conflict. On the ISA bus, which was the mainstream of the IBM compatible 16-bit personal computer at the time, it was necessary to manipulate a small DIP switch installed on the card so that the addresses of the connected cards do not collide. But such a procedure was not necessary in MSX.

In general, when peripheral devices are connected to the system bus, it is necessary to install driver software to handle the peripheral devices. Since the memory containing such a program can be connected to the slot of the MSX, the necessary software is accommodated in the cartridge of the hardware of the peripheral device. Therefore, merely inserting a cartridge of a peripheral device, such as a modem, into the slot, a driver software and an application software are simultaneously incorporated into the system. Hence, we could eliminate so-called installation work. Such a complete 'plug and play' function is an important one to allow children, who were the MSX main users, to use the system without any special computer skill.

The slot architecture also helped each manufacturer to add features to differentiate their products. For example, Yamaha had released a product incorporating music synthesiser hardware using the slot architecture. They released a synthesiser hardware together with a unit for the slot having an extended



Figure 11. FS-A1ST MSX computer from Panasonic (MSX Turbo R) (1990) with MIDI Interface cartridge inserted in the SLOT.

function of the BASIC language to program the synthesiser, and application software to use the synthesiser. Thus the slot architecture realises a mechanism that not only hardware but also software can be incorporated without changing MSX system software. Another example shown in Figure 11 is the Panasonic MSX computer released in 1990 in which a word processor software is incorporated using the slot mechanism.

6.3. MSX and kids

MSX can be purchased cheaply and easily, since it uses an ordinary television receiver for display and audio cassette tapes for external storage. Hence, not only adults but also many children became its users. In particular, many game applications made the machine friendly to children. However, unlike game-specific machines, MSX has a rich set of basic functions as a computer, including Japanese language support. MSX was optimally designed using the trial and error knowledge of developing the preceding 16-bit personal computers. The basic software of MSX is the BASIC language interpreter. When the user turns on the power, by default, the command input screen of the BASIC interpreter is displayed and enters in the standby state as shown in Figure 12. With this feature, children were naturally able to have opportunities to touch a programming language. By this, MSX could provide an entrance to nurture many computer engineers.

6.4. Growth of MSX

MSX is a computer developed mainly by ASCII Corp. Since ASCII originally started as a publishing company, it had the function of issuing books within the company. In order to promote MSX to be widely used, we released the internal information of the system as books. For example, assuming that the



Figure 12. Start screen of MSX computer.

internal module is called from the application software, we disclosed not only the entry addresses of various modules of OS and BASIC software but also the source code of the system software called BIOS that handles hardware operation. Also ASCII published many books to guide the use of MSX. As a result, many software and hardware were provided by third-party manufacturers, and it was widely accepted as a consumer-oriented personal computer.

However, MSX was a machine positioned between game machines and 16-bit personal computers, and thus sales gradually became stalled. Finally, it was decided to cease production and sales of MSX in 1995. However, it has been supported by enthusiastic users since then. There were moves to restore the system using software emulator running on a personal computer, and even now there are many users worldwide who play with MSX.

We had released four versions of the MSX system, i.e. MSX, MSX 2, MSX 2+, MSX turbo R in 7 years. But even the last version, MSX turbo R released 7 years later, was able to run the application software of the first MSX. The main users of MSX are children who touch the computer for the first time. We believed that we should not waste software, hardware, and know-hows developed in the previous versions, and thus we strived with high priority to maintain compatibility between different versions.

6.5. Internationalisation of MSX

MSX had been sold not only in Japan but also in the following countries, with the support of Microsoft.

- European countries such as the UK, France, the Netherlands, Germany, Spain, Italy, and Russia
- Korea
- Middle East Countries
- Brazil, Argentina, Chile, Uruguay.

In order to be used in these countries, it is necessary to support each national language. As for the languages in Europe and South America countries, basically support of character set and keyboard layout was sufficient. We developed versions of MSX supporting them with reference to the characters and keyboard layout of 16-bit personal computers. Since it was difficult to handle Arabic language which writes from right to left, we developed with the help of engineer from Al-Alamia who was Kuwait's distributor. A Korean version that supports Hangul characters was also developed with the cooperation of Korean manufacturers. Promotion of MSX overseas had some difficulties when compared with Japan, partly because publications from ASCII Corp. do not reach directly due to

language barriers. However, in some countries, MSX was accepted as the most popular computer for consumer.

7. Products that competed/coevolved with personal computers

In Japan, there were various products that competed/coevolved with personal computers. They are, for example, office computers, word processors, game machines, electronic organisers, etc. In this section, we see how they evolved and how they competed with personal computers.

7.1. Office computers

There are two major demands for computer use in the office: they are accounting and document preparation. Considering today's office software for personal computers, they are roughly corresponding to spreadsheet and word processing software. In Japan, besides the history of developing mainframe computers with the IBM architecture, there was another history of developing computers that have unique architectures and are called 'office computers.' Such a kind of computers were produced to meet the demands of small offices that cannot introduce expensive mainframe computers for computerisation.

While mainframe computers were developed by introducing U.S. technology, office computers were developed by the proprietary technology of each Japanese manufacturer from around 1961. Such office computers were provided with a package of application software and support services. However, as the performance of personal computers improves, office computers became developed on a global standard platform such as Windows that is cheaper than the ones with their own architectures. Finally, they were absorbed by personal computers and faded out.

7.2. Word processors

In the U.S.A. and Europe, typewriters were used for document creation in the office, but since the number of Japanese characters is very large, mechanical Japanese typewriters were not very popular due to the poor productivity. The Japanese word processor, in which characters can be input by keyboard, edited on screen, printed by printer to create documents, was first introduced by Toshiba in 1978. The price was 6.3 million yen which was not an affordable price, but it was recognised that certainly there is a market. New models were thrown into the market actively from each manufacturer one after another. As a result, products in the 160,000 yen range were introduced to the market in 1985, and miniaturisation also progressed. In this way, they were widely spread into small business offices and individuals.

Basically, a word processor is a dedicated computer equipped with a printer that executes word processing software as an embedded application. It had been widely accepted because of its easy-to-use design such as the quick start-up feature, and a keyboard layout focusing only on the word processing task (Figure 13).

These word processors had continued to progress. Besides the word processing function, they were equipped with spreadsheet and communication applications. And finally, more sophisticated machines than ordinary personal computers appeared. However, the format of the document file of the word processor dedicated machine was unique to each manufacturer. Hence, it was not easy to exchange documents between word processors from different makers. In addition, as for the communication function, the browser environment on the personal computer became better to use, since browsing the Internet became more important.

Meanwhile, with the rapid spread of personal computers, the number of users of word processing software on personal computers had increased, and cheap and high quality printers became available for them. Also, MS Word files were able to be used for exchanging documents as attachment to mail. Thus the dedicated word processor machine had declined (Figure 14).



Figure 13. Fujitsu OASYS-30AD Word Processor.

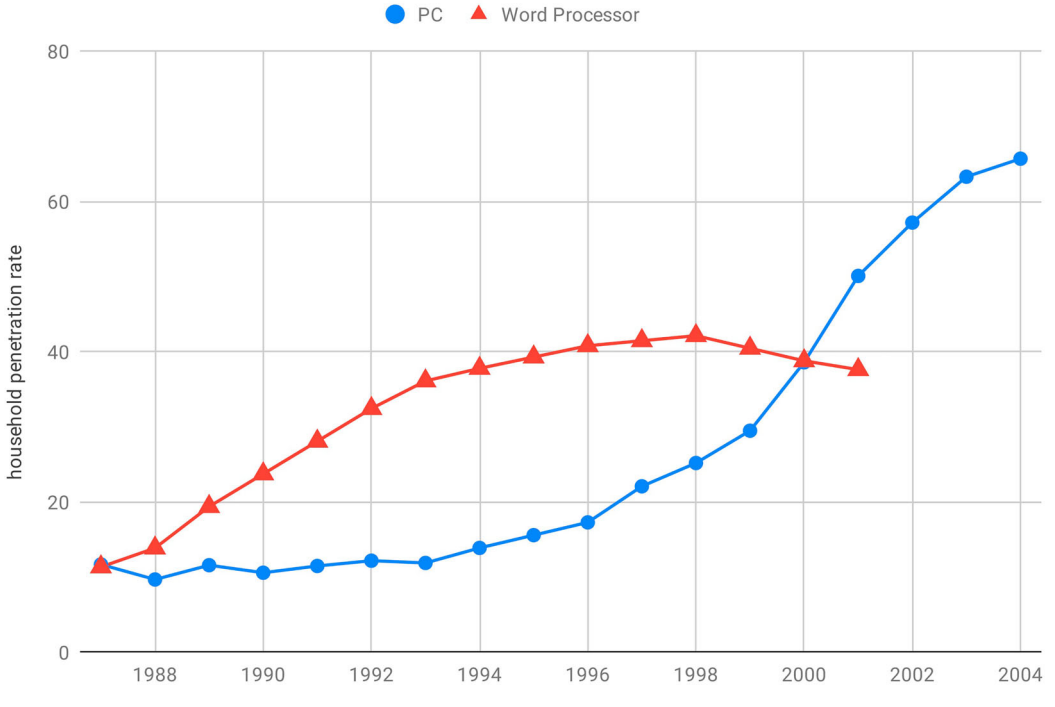


Figure 14. Household penetration rate of personal computer and word processor dedicated machine (source: Cabinet Office consumption trend survey).

7.3. Game machines

Game machines are computers specialised in playing game software, developed with emphasis on animation performance and audio function. Since the power of screen and music largely determines the attractiveness of the game contents, state-of-the-art semiconductor technology for graphics and audio had been introduced into the game machines.

The market structure of game machines is a closed one both for hardware and software, showing a completely different aspect from the market for personal computers where global standards have been developing. The objective of the manufacturers is to make the price of game machines as cheap as possible to lower the purchasing barriers for consumers. They compensate the production costs of game machines, by enclosing game software developers and ensuring royalty income from the sales of game software. To keep the mechanism to compensate the cost of the game machine by the sales of the software, no technical information of the game machine is disclosed. Hence, there is no room for compatible machine makers to come into the market. Recently, however, game makers who develop game contents for smartphones are increasing, since the performance of smartphones has improved, and playing attractive game contents is possible. Therefore, it is becoming difficult to maintain the old business structure of game machines.

Nintendo's Family Computer was the first successful game machine in Japan. Table 3 shows a list of major game machines released from Japanese game machine manufactures.

7.4. Electronic organiser

An electronic organiser was sold as a gadget that can electronically record and refer to schedules, telephone directories, memos, etc. for a businessperson as a target customer from around 1980. Zaurus, a typical electronic organiser released from Sharp, had acquired the largest market share. On the other hand, in 1996 Microsoft released the Windows CE operating system, allowing other companies to enter the market, and established a product genre called PDA (Personal Digital Assistant) (Figure 15).

Later, those with a communication facility were also sold, some of which may be considered as prototypes of the current smartphone. For example, Locatio, which was released from EPSON in 1999, equipped with a mobile phone, GPS and digital camera (Figure 16). Thus it had the basic functions of

Table 3. Release year of major game machines in Japan.

Year	Machine
1983	Nintendo Family Computer
1985	SEGA Mark III
1987	SEGA Master System, NEC PC Engine
1988	SEGA Mega Drive
1989	Nintendo Game Boy, NEC PC Engine Super Graphics
1990	Nintendo Super Family Computer, SEGA Game Gear, SNK NEOGEO
1991	NEC PC Engine Duo
1992	SEGA Wondermega
1994	Sony PlayStation, Panasonic 3DO, SEGA Saturn, SNK NEOGEO CD
1996	Nintendo 64
1998	SEGA Dreamcast
1999	BANDAI WonderSwan
2000	Sony PlayStation 2
2001	Microsoft XBox, Nintendo GAMECUBE
2004	Nintendo DS, Sony PlayStation Portable
2005	Microsoft Xbox 360
2006	Sony PlayStation 3, Nintendo Wii
2011	Nintendo 3DS, Sony PlayStation Vita
2012	Nintendo Wii U
2013	Sony PlayStation 4, Microsoft XBox One
2017	Nintendo Switch



Figure 15. Sharp Electronic Organiser PA-Z900 (1997).

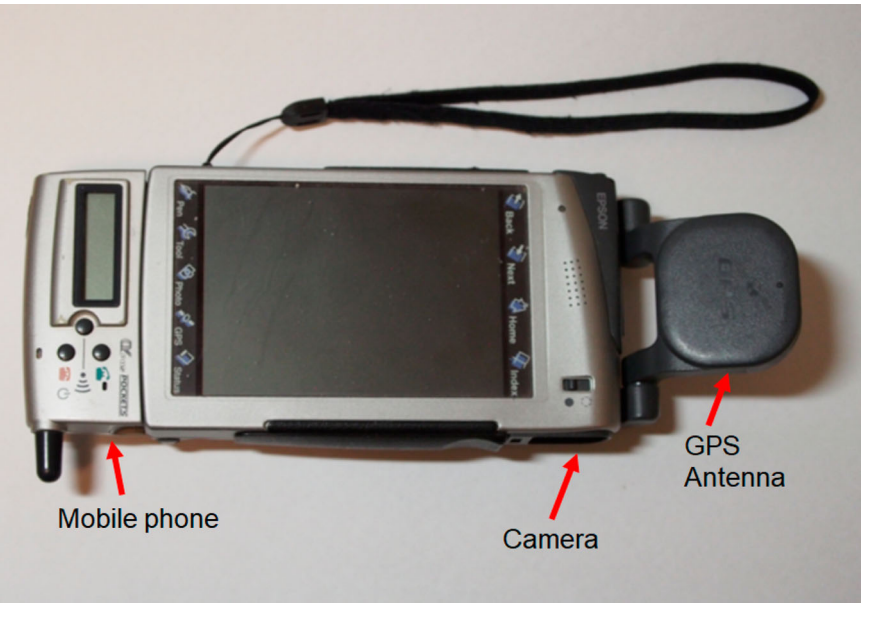


Figure 16. EPSON PNV-1000P Locatio PDA (1999).

the current smartphone. However, as the functions of mobile phones became abundant, products of this genre were absorbed by them.

8. 16-bit personal computers

NEC's PC-8000 series personal computer was the first one that acquired a large share in the Japanese personal computer market (see Section 4.1). It had an 8-bit CPU and was a computer of the same class as Apple II, Commodore PET, and Tandy TRS-80. It should be noted that PC-8000 series was manufactured and sold by the NEC's semiconductor division, though NEC was a manufacturer of a mainframe computer. Its reason seems that the early 8-bit personal computer was considered as a toy for hobbyist from the computer department who was manufacturing a large computer for business use. However, as it became possible to use a 16-bit CPU, it turned out that personal computers are usable for business. IBM was aware of it and entered into this market in 1981. Under such circumstance, the section which was in charge of NEC's office computer developed and released the PC-9800 series personal computer in 1982, an upward compatible machine with PC-8000 series.

8.1. NEC PC-9801 and AX architecture

The PC-9800 series was equipped with a BASIC language processing system compatible with the PC-8000 series in order to inherit the business application software created with the BASIC language of the PC-8000 series. Furthermore, it was able to inherit assets such as know-how of software makers cultivated in the BASIC language processing system of the PC-8000 series. The PC-9800 series was able to acquire the largest share in the Japanese business oriented personal computer market. Another reason why it acquired a large market share is that it was able to display kanji characters at high speed with hardware support.

On the other hand, other personal computer makers established an industry organisation called AX consortium for jointly promoting personal computers based on the AX standard[15] advocated by ASCII Corp. and Microsoft in 1987 and based on AX standard they developed and marketed product called AX computers. The AX standard was a specification of a personal computer developed based on the global standard IBM PC/AT architecture and supported high-speed kanji character display using a special graphic chip developed by ASCII Corp. The AX standard was a plan formulated with the idea inspired by the successful MSX standard. However, since Fujitsu, Toshiba and Panasonic sold personal computers with its own architecture incompatible with the AX standard, supports from software makers were limited and NEC 's monopoly became more robust.

After that, in 1987 IBM introduced the graphic hardware called VGA which had better vertical resolution, and it became the standard resolution for personal computers. However, at that time the performance of the CPU improved, and kanji character rendering with software alone became practical without a help of dedicated graphic processor. In 1990, IBM Japan released a PC specification so-called DOS/V, in which software renders kanji characters on VGA. This specification became the standard architecture of Japanese personal computer later on. In order to promote DOS/V, an industry organisation called OADG (The PC Open Architecture Developer's Group) was established under the initiative of IBM Japan in 1991. OADG tried to realise the application platform for the Japanese personal computer based on the IBM PC/AT architecture, which became the world standard against the NEC's PC-9800 series which dominated the Japanese market. This attempt became successful by making it possible to provide personal computers for Japanese market with inexpensive PC/AT compatible hardware by rendering kanji characters with practical performance only with software.

8.2. Widespread of Windows and decay of PC-9801

NEC's PC-9800 series personal computer, which had acquired a large market share due to its high-performance kanji character display capability, had become less favourable due to the popularisation

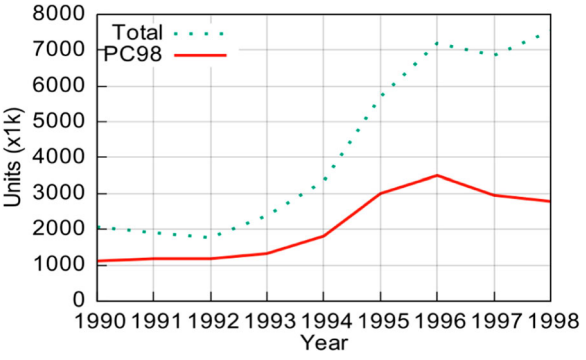


Figure 17. Rise and Fall of NEC PC-9800 Series Personal Computer. It shows the NEC PC-9800 series and Japanese PC domestic shipments from 1990 to 1998. Data collected by NEC and JEITA (source: https://commons.wikimedia.org/wiki/File:PC98_and_domestic_PC_shipments_in_Japan_1990-1998.svgCC4.0).

of DOS/V and Microsoft' s Windows operating system. As Windows, which employs bitmap graphics, became popular, the superiority of the high-speed kanji character display graphic architecture of the PC-9800 series was impaired. Application software developers were now able to run on any manufacturer's personal computer at a practical speed by developing applications on Windows graphics architecture, because a high-performance CPU was available. The PC-9800 had locked in software developers to its unique architecture, but its structure became difficult to maintain (Figure 17).

In this way, the architecture of the Japanese personal computer was consolidated into the global standard architecture of DOS/V which is based on IBM PC/AT architecture and Windows. The PC-9800 series personal computer, which had a proprietary architecture, was able to exist protected by the hardware barrier of the kanji character display performance. It finally ceased accepting orders in 2003 after finishing their role.

8.3. IBM compatible chipset

During this period, there was a development of technology that made it easy for manufacturers other than IBM to produce IBM compatible machines. It is a series of LSI called PC compatible chipset. The first chipset was supplied by a U.S. company called Chips & Technologies[16] (C&T), the first fabless semiconductor maker founded in 1984. Mitsui & Co., Yamaha, Kyocera, ASCII Corporation invested and helped the establishment of this company by ASCII's leadership.

With this chipset, IBM compatible machines can be easily and inexpensively manufactured, and many cheap IBM compatible machines were supplied to the market. ASCII acted as a distributor of Chips & Technologies in Japan. Also, for AX standard personal computer competing with the top share NEC PC-9800 series personal computer, ASCII developed and distributed a graphic processor called JEGA in collaboration with C&T to display kanji characters at a high speed.

9. Embedded computers

Unlike computers operated by people equipped with a keyboard and a display, an embedded computer is a computer that is built into an apparatus and controls it. The biggest difference from a general computer such as a personal computer is the following: it is embedded in products produced in very large quantities, and thus it is required to have low price, low power consumption and small size. There are also high demands for real-time performance and high reliability because when a problem occurs, it cannot expect any help by human, and thus it needs to keep working without intervention.

Table 4. Japanese supercomputers ranked within 10th place on the TOP 500.

	Rank	Computer	Manufacturer	GFLOPS
1989	1	SX-3/44R	NEC	26
1992	5	VPP500/30	Fujitsu	48
1992	10	S-3800/480	Hitachi	32
1993	1	Numerical Wind Tunnel	Fujitsu	280
1996	1	SR2201/1024	Hitachi	307
1997	2	CP-PACS/2048	Hitachi	614
1999	4	SR8000/128	Hitachi	1024
2000	6	SR8000-F1/100	Hitachi	1200
2001	5	SR8000/MPP	Hitachi	2074
2002	1	Earth-Simulator	NEC	40,960
2003	7	PRIMEPOWER HPC2500	Fujitsu	11,980
2004	7	RIKEN Super Combined Cluster	Fujitsu	40,960
2006	7	TSUBAME/Sun Fire X4600 Cluster	NEC/Sun	49,869
2011	1	K computer	Fujitsu	8,773,630
2017	7	PRIMERGY CX1640 M1	Fujitsu	24,913,459

Real-time feature becomes especially important when controlling mechanically operating machines such as printers or car engines. In order to control the device in real time, it is necessary to have a special OS called a real-time OS having a management function that guarantees that the processing of individual tasks is definitely completed within a certain time. Initially, operating systems developed individually by each manufacturer were used, but as the functions required by the OS became more sophisticated, the OS provided by real-time OS vendor was widespread.

Such an embedded system is now becoming important in the field called IoT (Internet of Things), as a component constituting a network to which everything is connected to. The TRON project proposed by Ken Sakamura in 1984 is worthy of attention as a move to realise the unified architecture covering the application field of today's computer. The TRON project tried to release the developed system free of charge and to provide the infrastructure resources required by a computerised society. The base real-time OS is ITRON[17] (Industrial TRON), which was used by many manufacturers. In order to secure the safety of the networked system, it is desirable to have a publicly disclosed operating system with security ensured widely. TRON can be a system that satisfies that requirement.

10. Supercomputers

Unlike mainframe computers and personal computers which form mass-produced software and hardware markets, supercomputers do not form large markets. So, it is not likely to become a market that is monopolised by a small number of companies. There is a pure demand for a high cost-performance computer manufacturing capability. Japanese manufacturers are good at this field who have high technical ability to develop computers and are able to manufacture the world's highest level supercomputers. Table 4 is a Japanese supercomputer ranked within 10th place on the TOP 500 site.[18]

11. Concluding remarks

When new technologies such as mainframe computers or personal computers were introduced to the market, many manufacturers competed each other because everybody had a chance to be the market leader who gets the largest market share. But, over time, the market mechanism works and only a few systems supported by third party manufacturers could survive. For this reason, pioneers in the market will focus on first obtaining the largest market share. However, when the market matures to a certain extent, it becomes clear who the market leader is that occupies the largest market share. It is time for

other manufacturers to switch to a strategy to develop products that make use of their strengths while having compatibility with market leader products, for survival.

However, particularly technically superior manufacturers have a desire to commercialise their own ideas and succeed, it is difficult to switch their strategy because developing the same product as market leader will impair their pride. Fujitsu was able to make this difficult decision to develop products superior to market leader's products and they were able to succeed in the IBM plug compatible product market. In the personal computer industry, many overseas manufacturers were able to build a successful business by entering the market with IBM compatible machines.

Unlike overseas, NEC's PC-9800 series personal computer once monopolised the Japanese market with proprietary hardware that assisted kanji character display function, because IBM compatible machines had the lack of graphics performance for displaying kanji characters. Since NEC did not approve other manufactures to produce PC-9800 compatible machines, they were able to monopolise the profit as a computer manufacture. However, due to the improved performance of the CPU, proprietary hardware for displaying kanji characters became no longer necessary, and the market has been standardised by OADG machines of IBM compatible architecture.

In the history of Japanese computers, there was a period when Japanese unique systems became popular due to the barrier of Japanese language support, but ultimately the language barriers were overcome by improved performance of semiconductor, and they were integrated into a global competitive market and Japan's original system disappeared. For the hardware that gained the largest market share, its manufacturing costs are reduced due to mass production effects, so it gets further increasing market share. This effect with extreme improvement of the semiconductor performance was particularly remarkable in the personal computer. By this, not only the mainframe computer but also the office computer and the dedicated word processor, which were original product of Japan, lost their market.

Meanwhile, in the mobile field, mobile phones built a very large market, and the PDA market was absorbed by it. There were many machines with proprietary architecture in Japan, but in the era of smartphone, global standard machines are sweeping the market. In this way, Japanese computers in many fields have been forced to do the business in the global competitive market even inside of Japan.

In the industries such as home appliances, cars and cameras, high quality products made in Japan have gained a large market share in the world market, but this phenomenon did not occur in the computer industry except for M100/M10/PC-8201 and MSX. The reason for this difference is, as I mentioned at the beginning, unlike the products mentioned above, it needs software and peripheral equipment from third-party manufacturers to live, since the computer cannot survive only with the computer itself. This is because it is a market that consists of many stakeholders which is not easy to build.

In the game machine industry where such global market principle does not work, the system in which Japanese unique architecture is spreading in the world market is still maintained. However, due to the improved performance of semiconductors, it is now becoming possible to develop games that can be enjoyed on open architecture smartphones, and it will be difficult to expand the market beyond current size for dedicated game machines.

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Disclosure statement

No potential conflict of interest was reported by the author.

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